

FOL: Bridging Object-Oriented and Functional Programming via the Metaobject Protocol

Frank Adrian
Ancar Technology
Olathe, KS, USA

frank.adrian314159@gmail.com

Abstract

Clojure [16] omits CLOS; Hickey [17] argues that mutable objects are a complexity hazard incompatible with immutability rather than the power boost CLOS offers. In FOL (Functional Object Lisp), we use the Metaobject Protocol (MOP) to back instances with persistent structures, preserving CLOS's open dispatch model while enforcing immutable reference semantics. FOL contributes a persistent metaclass with hybrid storage and lazy migration, and a specificity-ordered predicate dispatch model.

FOL's slot reads incur 1.4–2.2× overhead; slot updates cost 40–250× in isolation. At idiomatic workload scales, FOL is 5–44× slower than mutable CLOS; for direct translation of fine-grained mutable algorithms (e.g., AST rewriting), overhead reaches 202×. However, at extreme scale, persistent state maps avoid memory fragmentation that degrades the mutable baseline, yielding a 3.8× runtime advantage in our logic simulation benchmark (Section 5.4). Predicate dispatch uses a level-based ordering sorted at each method-definition in $O(ND \log N)$ (D = hierarchy depth), avoiding the pairwise theorem-prover overhead of subsumption systems; the trade-off is an $O(N)$ per-call linear scan (vs. $O(1)$ cached dispatch), with closed defn dispatch indistinguishable from hand-written cond cascades ($\approx 2.0\times$).

CCS Concepts

• **Software and its engineering** → **Constraints; Classes and objects; Functional languages; Object oriented languages; Extensible languages.**

Keywords

Functional programming, object-oriented programming, Lisp, persistent data structures, CLOS, MOP

1 Introduction

CLOS is extensible, but mutable objects create three problems for speculative workloads, evolving schemas, or value-dispatched guards:

P1 — Costly structural sharing. CLOS lacks native structural sharing; speculative workloads must copy objects in $O(N)$ in the number of slots for every snapshot. This makes fine-grained snapshotting prohibitively expensive compared to persistent alternatives (FSet [5], Sycamore [9]), which nonetheless lack CLOS integration. FOL's isolated slot updates cost 40–250×, but $O(\log n)$ structural sharing bounds per-update cost; at sufficient scale, persistent instance state maps avoid memory fragmentation that degrades mutable baselines (Section 5.4).

P2 — Schema evolution labor. Renaming slots requires manual update-instance-for-redefined-class methods. FOL's `alias`

annotation recovers renamed slots at first access, composing multi-hop chains in a single pass (Section 5.1.2).

P3 — Inexpressive predicate dispatch.; value-level conditions CLOS dispatch is type- and value-limited. filtered-functions [24] extends this via subsumption but requires expensive pairwise logical-implication checks over Boolean predicates. FOL's syntactic ordering is decidable at load time in $O(ND \log N)$; per-call overhead approaches manual cond cascades ($\approx 2.0\times$; Section 5.1.1).

FOL uses the MOP [19] to back instances with persistent structures and adds specificity-ordered predicate dispatch. Slot mutation becomes assoc, while other CLOS interfaces remain (Table 1). The contributions compose: Listing 4 gates on values while reading persistent state, while Listing 3 enables $O(1)$ update discard without copying.¹

Contributions:

- (1) A **persistent metaclass system** (Sec. 2) with hybrid native/trie storage and lazy alias-based migration (Sec. 4.2).
- (2) A **specificity-ordered predicate dispatch** (Sec. 4) using load-time decidable leveling without theorem provers, integrated into CLOS.

2 Architecture

Slot storage uses a hybrid layout: each persistent object has $T = 32$ native positions, 6 reserved for system overhead (trie pointer, schema versioning, metadata), leaving 26 for user slots. Most classes fit this layout [20]; we confirmed this in Common Lisp via a survey of 295 Quicklisp definitions where 97% of classes fell within this limit. Redefinitions are lazy: renames resolve via alias maps; new slots acquire initforms at first mutation. Redefinitions exceeding 26 slots trigger trie conversion.

persistent-class enforces immutability via two-level mutation blocking: FOL's transpiler rejects (`setf slot-value`) at compile time as a syntax error for FOL code; direct CL code calling (`setf (slot-value ...)`) is caught at runtime via an MOP override that signals an error. By dropping `change-class`, FOL replaces state-machine identity with functional projection, borrowing CLOS syntax and MOP mechanics. `assoc` returns a fresh instance sharing nodes via path copying.

2.1 Collections and Runtime

FOL implements persistent collections via 32-way tries [3, 23]. FOL transpiles to Common Lisp via a two-stage pipeline: (1) the *Pattern Reader* parses extended syntax, and (2) the *Macro Expander* emits

¹FOL addresses the Expression Problem [27]: new functions and clauses can be added over existing classes without modifying original code.

standard CL. This leverages the host compiler’s optimizations while maintaining FOL’s semantics.

2.2 Space Complexity

Identity types hold current values; unreferenced snapshots are GC-eligible. Each assoc copies one root-to-leaf path ($O(\log_{32} N)$ nodes), bounding allocation by depth rather than size. SBCL nursery reclamation keeps GC overhead at $\approx 1.5\%$ in the LSim benchmark (profiled at the 32x32x32-30000 configuration via sb-sprof).

3 Features and Dispatch Model

FOL adopts Clojure’s literal syntax and sequence APIs. To ensure interoperability, FOL shadows `cl:assoc` with a generic function; within the `fol` package, `assoc` dispatches to persistent structures, while remaining accessible to standard lists via a fallback method.

Concurrency and State Management. FOL provides transactional refs, atoms with compare-and-swap, and asynchronous agents as direct ports of Clojure’s concurrency model. These are verified against Clojure’s original test suites and inherit their semantics. Transducers and lazy sequences are similarly ported; transient builders reduce bulk allocation overhead by amortizing copy costs during construction.

Dispatch and Method Combination. Generic functions dispatch by arity, then specificity (Section 4). `textbfClosed` dispatch (`defn`) emits a static cond cascade; **open dispatch** (`defmethod`) supports CLOS method combinations (`:before`, `:after`, `:around`) and extensibility via `call-next-method`. Table 2 summarises the criteria. The two forms are mutually exclusive per function name: `defn` emits `defun` and `defmethod` emits `defmethod`.

4 Predicate Dispatch

Figure 1 formalizes FOL’s multi-pattern dispatch syntax.

Note that *s-clause* with embedded map destructuring is the idiomatic equivalent of *m-clause*, which serves as a shorthand.

4.1 Dispatch Ordering Rules

FOL’s predicate specializers (`(var (fn args...))`) evaluate at dispatch time; predicates are more specific than types. Specificity categories (Table 3) establish a partial order; incomparable patterns at the same level resolve by explicit preference registry (see Section 4) or definition index. $\prec_{\text{disp}}$ conservatively approximates semantic specificity: Level 3 $>$ 2 $>$ 1 $>$ 0. Level 2 uses semantic subclassing (Spec-Type); multi-parameter patterns apply Spec-Prefix and Spec-Tail position by position. Level 1 sets fixed-vs-rest precedence (Spec-Fixed). Consider:

```
(defn classify
  ([x <integer>] :any-integer) ; catch all
  ([x (even?)] :even))       ; special case
```

FOL reorders this to `even?` first, so even integers never reach `:any-integer`. User-defined predicates must be pure and total; since the dispatch scan has no condition-handler boundary, conditions signaled inside a predicate propagate directly to the caller. Closed `defn` groupings trigger compilation warnings for shadowed clauses. Definition-order reliance creates modularity hazards; we use explicit mechanisms like `prefer-method` to mitigate this issue.

Method qualifiers execute in CLOS-standard order (e.g., most-specific-first `:before`), but $\prec_{\text{disp}}$ replaces the CPL. This deliberately diverges: predicates execute before types, prioritizing behavioral constraints over structural inheritance.

Conjunction takes the max level of its components; two Level-3 patterns like `(and (even?) (prime?))` and `(even?)` are incomparable and ordered by preference registry or definition index. Disjunction merges domains (less specific than either conjunct), so or patterns are atomic Level-3 and ordered by tiebreaker. Negation is excluded: (not *p*) ordering is undecidable, so negations fall to the tiebreaker.

Type-Annotated Conjunctions: Path to Resolution. `(and (<integer>) (prime?))` receives Level 3 ($\max(2, 3) = 3$), the same level as the bare predicate `(prime?)` alone, even though the conjunction is strictly more constrained. The current syntactic calculus cannot determine this semantic subsumption natively, requiring definition order as a modularity hazard. However, a straightforward compile-time enhancement—a *composite level* (e.g., Level 2.5)—recovers structural subsumption with minimal overhead. At load time, a linear scan partitions conjuncts by their constituent levels; conjunctions combining a type (Level 2) with a predicate (Level 3) are assigned to the composite level, ordering them before bare predicates. This requires no runtime cost: only a single pass across the pattern’s conjuncts at method-definition time. This approach is implementable and planned for a near-term release, eliminating the deficiency while preserving the current level system’s simplicity.

4.1.1 Formal Specificity Ordering. We formalize the specificity relation $\prec$ using the inference rules summarized in Table 4. Let $\mathcal{L}(p) \in \{0, 1, 2, 3\}$ be the specificity level of pattern *p*. **Syntactic form classes:** wildcard (*x*, *y*), type annotation (*<integer>*, *<string>*), predicate (`(even?)`, `(> 10)`), sequential (`[x y]`, `[x (> y 5)]`), and map (`{:key x}`, `{:a (number?)}`). Maps are sorted canonically via `string<` on symbol-names. $p \sim q$ iff both have the same level and form class. For compound patterns, $\mathcal{L}([p_1, \dots, p_n]) = \max_i \mathcal{L}(p_i)$ and $\mathcal{L}(\{k_1 p_1, \dots\}) = \max_i \mathcal{L}(p_i)$: the level is the max element level. Spec-Level applies to compound patterns globally: `[x (even?)]` (L=3) fires before `[x <integer>]` (L=2). Positional rules (Spec-Prefix, Spec-Tail) apply only when compound patterns share the max level ($p \sim q$); Spec-Level takes precedence otherwise. **Convention:** $p_1 \prec p_2$ means *p*₁ fires before *p*₂—i.e., *p*₁ is more specific. Higher levels are more specific, so Spec-Level concludes $p_1 \prec p_2$ when $\mathcal{L}(p_1) > \mathcal{L}(p_2)$.

Rule priority: Spec-Level takes precedence. Within a level, Spec-Type refines Level-2 patterns via class hierarchy; sibling classes fall back to preference registry or definition index. Spec-Prefix and Spec-Tail apply lexicographically after Spec-Level partitions by max level.

For `defmethod`, the dispatch sort $\prec_{\text{disp}}$ re-sorts at each method-definition event, including REPL re-evaluation; for `defn`, the ordering is fixed at compile time. Let *idx*(*p*) denote the definition index of the method clause containing *p*. $\prec_{\text{disp}}$ is defined by the comparator: (1) $\mathcal{L}(p)$ descending; (2) if same level and $p_1 \prec p_2$, then *p*₁ sorts first; (3) if same level and incomparable under $\prec$, the preference registry takes precedence; if no preference exists, *idx*(*p*) descending breaks the tie. The partial order $\prec$ is applied strictly within levels; *idx* serves as the final tiebreaker.

Table 1: MOP Protocol Adaptation

Protocol	Status	Rationale
slot-value-using-class (setf slot-value-using-class)	Redirected Guarded	Reads from base object or overflow trie Rejects mutation at compile time for FOL code (transpiler error); runtime error for direct CL calls via MOP override
slot-boundp-using-class	Redirected	Checks object slot membership
slot-makunbound-using-class	Blocked	Immutability invariant — use dissoc to unbind slots
validate-superclass	Extended	Cross-metaclass mixing
initialize-instance	Extended	Populates base object and overflow trie from initargs; stamps %schema-version (an integer matching the class’s current version counter) at birth, enabling update-instance-for-redefined-class to detect stale instances
reinitialize-instance	Extended (:before on metaclass)	Snapshots current alias-map, overflow indices, and version counter into a class-version struct before each redefinition; advances the version counter
compute-slots	Specialized	Implements hybrid native/trie layout
update-instance-for-redefined-class	Implemented (:after)	Multi-hop chain replay: composes alias-maps from version chain back to birth version, recovering renamed values (Sec. 4.2)
change-class	Omitted	Changing a live object’s class violates immutability; callers construct a new instance of the target class instead. Internal SBCL redefinition triggers are handled by constructing a fresh instance from the old state during the next access.

Method combination uses standard CLOS execution semantics (:before/:after/:around). $\prec_{disp}$ replaces the class precedence list; as a total order, it preserves CLOS’s most/least-specific-first guarantees; definition index serves as the final tiebreaker that ensures totality.

```

defn    ::= (defn name (s-clause | m-clause)) | (fn (s-clause | m-clause))
defg/m ::= (defgeneric name...) | (defmethod name qualifier2 (s-clause | m-clause))
s-clause ::= [param*] body+
m-clause ::= {(symbol param | :keys [(symbol | (sym param))*])*} body+
param    ::= symbol | (symbol type) | (symbol (fn arg*)) | destr
destr    ::= [param*] | {:keys [(symbol | (sym param))*]}
type     ::= <type-name>
qualifier ::= :before | :after | :around

```

Figure 1: FOL Multi-Pattern Dispatch Grammar (compressed)**Table 2: Choosing defn vs. defmethod**

Requirement	defn	defmethod
Downstream code may add clauses	×	✓
call-next-method needed	×	✓
:before/:after/:around qualifiers	×	✓
Minimum dispatch overhead (static cond)	✓	×

Exact-arity patterns of different lengths are incomparable; dispatch isolates the matching arity group first. For n arguments, this group contains all length- n fixed-arity patterns and length- $\leq n$ rest-accepting patterns. $\prec_{disp}$ is total over this group (Lemma 1). Spec-Fixed resolves the overlap by ordering the fixed-arity pattern

Table 3: Predicate Specificity Categories

Level	Category	Example
3	Predicates	(x (even?))
2	Types	(x <integer>)
1	Fixed-arity (all-wildcard params)	[x y]
0	Wildcard	x

as more specific. Preference-based and definition-order tie-breaking aligns with Lisp’s interactive workflow but introduces the modularity hazards discussed in Section ??.

LEMMA 4.1 ($\prec$ IS A STRICT PARTIAL ORDER; $\prec_{DISP}$ IS A STRICT TOTAL ORDER). *The syntactic specificity relation $\prec$ defined by Table 4 is a*

Table 4: Specificity Inference Rules

Rule	Formalization
Spec-Level	$\frac{\mathcal{L}(p_1) > \mathcal{L}(p_2)}{p_1 \prec p_2}$
Spec-Type	$\frac{C_1 \sqsubset C_2}{(x\ C_1) \prec (x\ C_2)}$
Spec-Prefix	$\frac{p_1 \prec q_1 \quad \mathcal{L}([p_1, \dots]) = \mathcal{L}([q_1, \dots])}{[p_1, \dots] \prec [q_1, \dots]}$
Spec-Tail	$\frac{p_1 \sim q_1 \quad [p_2, \dots] \prec [q_2, \dots]}{[p_1, \dots, p_n] \prec [q_1, \dots, q_n]}$
Spec-Fixed	$\frac{\mathcal{L}(p_i) = \mathcal{L}(q_i) \ \forall i \in \{1 \dots n\}}{[p_1, \dots, p_n] \prec [q_1, \dots, q_n \ \& \ r]}$
Spec-Map-Prefix	$\frac{p_1 \prec q_1 \quad \mathcal{L}(\{k_1\ p_1, \dots\}) = \mathcal{L}(\{k_1\ q_1, \dots\})}{\{k_1\ p_1, \dots\} \prec \{k_1\ q_1, \dots\}}$
Spec-Map-Tail	$\frac{p_1 \sim q_1 \quad \{k_2 \dots\} \prec \{k_2 \dots\}}{\{k_1\ p_1, k_2\ p_2, \dots\} \prec \{k_1\ q_1, k_2\ q_2, \dots\}}$
Spec-Keys	$\frac{\text{keys}(q) \sqsubset \text{keys}(p)}{p \prec q}$

strict partial order on arity-group patterns. The dispatch comparator $\prec_{disp}$ —which extends $\prec$ with the preference registry and the definition-index tiebreaker idx —is a strict total order on arity-group patterns.

PROOF. $\prec$ is a strict partial order. Irreflexivity: every rule requires a strict inequality (level, element position, key set, or subclass) that is false when $p_1 = p_2$. Transitivity: within a level the rules encode a lexicographic comparison on position, arity, and key set. For sequential patterns, transitivity is proven by induction on the sequence length n : Spec-Tail applies only when both sequences have length ≥ 2 ; inductive step by Spec-Prefix when heads are strictly ordered, Spec-Tail when heads satisfy $p_1 \sim q_1$ with the inductive hypothesis on the remainder. If sequences are length 1 with $p_1 \sim q_1$, or if heads differ in form, they are incomparable under $\prec$. For map patterns: Spec-Map-Prefix and Spec-Map-Tail follow the same lexicographic argument over key-sorted position; Spec-Keys follows from transitivity of strict set containment ($\sqsubset$). Across levels, irreflexivity ensures at least one strict level inequality along any $p_1 \prec p_2 \prec p_3$ chain, so Spec-Level or within-level rules give $p_1 \prec p_3$.

$\prec_{disp}$ is a strict total order. Incomparability under $\prec$ arises in four cases: (i) patterns differ only in variable names (same form, same level, no positional rule applies); (ii) sequential and map patterns share an arity; (iii) two Level-3 predicates are syntactically distinct with no positional relationship; (iv) two Level-2 patterns specialize on sibling classes. In each case the preference registry and tiebreaker $idx(p)$ supply a unique total rank; since every live clause carries a unique idx , no two clauses compare equal under $\prec_{disp}$. Cases (ii)–(iv) are modularity hazards: without an explicit prefer-method declaration, ordering depends on ASDF load sequence. $\square$

Predicate dispatch requires an $O(N)$ scan. FOL re-sorts at definition in $O(ND \log N)$. Subsumption systems achieve $O(1)$ dispatch via expensive provers; FOL trades runtime caching for syntactic tractability.

4.1.2 Dispatch Examples. Listing 1 shows a two-clause defn transpiled to a defun with a static cond cascade.

Listing 1: Transpilation: specificity-ordered multi-clause defn

```
;; FOL source: type specializer (Level 2) before wildcard (Level 0)
(defn classify
  ([x <integer>] :integer)
  ([x] :other))

;; Generated Common Lisp: ordering preserved as cond cascade
;; (class references bound once via load-time-value)
(defun classify (x)
  (cond
    ((typep x (load-time-value (find-class '<integer>))) :integer)
    (t :other)))
```

Comparison to defmulti. Clojure’s defmulti [16] requires manual resolution via prefer-method; FOL’s total order eliminates ambiguity for patterns with distinct types or structures.

4.1.3 Usage Patterns. Four patterns illustrate these features: structural diff (Listing 2) tracks nested changes by intercepting updates; validated update guard (Listing 3) enforces invariants; predicate dispatch on persistent state (Listing 4) gates behavior based on current values; and content routing (Listing 5) enables value-based alerting. In Listing 2, the :around method detects updates and stamps a version counter; the guard (= key :_changes) is essential to break the recursion that would otherwise occur when the method itself calls assoc to store the change metadata. In Listing 5, the pattern { :keys [(temp >= 100)] } illustrates the power of map-based predicate dispatch: the method only fires if the :temp key is present and its value satisfies the predicate, allowing for declarative, behavior-based event routing.

Listing 2: Automatic Structural Diff (FOL)

```
(defmethod assoc :around [(obj <diffable>) key val]
  (if (= key :_changes)
    (call-next-method)
    ; key=:_changes: breaks cycle -- calls assoc
    ; directly, skipping the branch that emits
    ; (assoc ... :_changes ...) and re-invokes :around
    (bind [old (get obj key)
           new-obj (call-next-method)]
      (if (= old (get new-obj key))
        new-obj
        (assoc new-obj :_changes (inc (get obj :_changes 0)))))))
```

Listing 3: Validated Update Guard (FOL)

```
(defmethod assoc :around [(obj <guarded>) key val]
  (if (valid-update? obj key val)
    (call-next-method)
    obj)) ; discard invalid update; original snapshot unchanged
```

Listing 4: Predicate Dispatch on Persistent State (FOL)

```
;; Level-3 predicate (critical?) tests the reading value at dispatch
time;
;; the body reads persistent slot :alert-history from the persistent
sensor
```

```
;; object without copying. Both contributions are required:
  predicate
;; dispatch gates the clause; persistent immutability makes prior
  history
;; a first-class value.
(defmethod record-reading :around
  [(sensor <sensor>) (reading (critical?))])
(bind [history (get sensor :alert-history [])]
  result (call-next-method))
(assoc result :alert-history (conj history reading))))
```

Listing 5: Value-Based Alerting (FOL)

```
(defmethod notify [{:keys [(temp (>= 100))]]]
  (print "ALERT: Critical Temperature!"))
```

Predicate forms evaluate their arguments dynamically at each dispatch call, responding to live variable updates (e.g., `*limit*`). However, mutating state inside the predicate closure corrupts dispatch correctness: the dispatch sort (compile-time for defn, load-time for defmethod) relies on stable predicate structures.

FOL’s validated guard avoids the $O(N)$ copy required in CL (Listing 6). While Clojure can achieve per-type guards using multi-methods to emulate `:around`, doing so forces all clients of standard update! to migrate to the new multimethod; FOL’s `:around` integrates securely beneath the existing `assoc` generic function without restructuring the hierarchy.

Listing 6: Validated Update Guard in Common Lisp — requires explicit copy

```
(defmethod update-obj :around ((obj guarded) key val)
  ;; No persistent objects: must copy before calling primary to preserve
  ;; original for potential discard. Cost:  $O(N)$  in number of slots.
  (let ((saved (make-instance (class-of obj)
    :slot-a (slot-value obj 'slot-a)
    :slot-b (slot-value obj 'slot-b))))
    (if (valid-update-p obj key val)
      (call-next-method)
      saved)))
```

FOL centralizes the guard and discards updates in $O(1)$ via persistence.

(1) The linear scan evaluates Level-3 predicates before type guards; they must be pure and total. (2) Load-order sensitivity affects incomparable patterns (e.g., sibling classes). FOL provides `prefer-method` to resolve these conflicts. (3) Detecting unreachable clauses is future work.

4.2 Schema Evolution

Chain replay lazily composes alias maps, recovering renamed values in one pass. Migration confluence prohibits slot names appearing as both source and target, preventing cycles. This is enforced via a DFS within the `reinitialize-instance` MOP protocol.

Listing 7: Lazy Schema Migration with Aliasing

```
;; Schema 1
(defclass <sensor> [] [[id] [reading]])
;; Schema 2: rename reading -> val; :reading keyword still routes to val
(defclass <sensor> [] [[id] [val :alias reading]])
```

At first post-redefinition *mutation*, chain replay writes `:reading`’s value into `:val`; reads of `:reading` before that mutation resolve via the alias map without modifying the instance. Cost is bounded at 5.4–12.2 μ s/instance across 1–3 hops (Section 5.1.2).

5 Evaluation

FOL fills a gap between CLOS and Clojure (Table 5), providing MOP-integrated persistence and full CLOS method combinations.

Table 5: Built-in Feature Comparison

Feature	FOL	Clojure	CL
Static type safety	×	×	×
Persistent objects	✓ ^d	o ^c	×
CLOS/MOP	✓	×	✓
Method combinations	✓	×	✓
Predicate dispatch	✓	single ^a	×
Schema evolution	✓ ^e	×	Manual
Transducers	✓	✓	×
Macros	✓	✓	✓
Raw scalar performance	Low ^f	High	High

^a `defmulti` dispatches via a user-supplied function; `prefer-method/derive` allow manual overrides but there is no automatic multi-parameter specificity ordering.

^b Multi-parameter type dispatch natively; libraries like `trivia` add pattern matching.

^c Clojure has persistent *data structures*; `defrecord` is map-backed, not MOP-integrated.

^d Slot storage backed by immutable HAMTs; (`setf slot-value`) blocked; updates via `assoc`.

^e Renamed slots recovered automatically at first post-redefinition access; multi-hop chains composed in a single pass.

^f Scalar arithmetic and non-slot operations inherit full CL performance. “Low” refers specifically to persistent object slot access: reads add 1.4–2.2 $\times$ overhead and writes add 40–250 $\times$ overhead relative to mutable CLOS.

5.1 Micro-Benchmarks

Persistent objects vs. mutable CLOS: **Construction** is 5 $\times$ native via `direct-allocate-instance/initialize-instance`. **Reads** add 1.4–2.2 $\times$; **Updates** are 40–250 $\times$ slower. At $T = 32$, $\approx 97\%$ of classes fit in native slots; overflow uses tries only when needed.

Table 6: Actual slot update overhead (μ s/op) vs. native CLOS

Slots	Persist.	Mut.	Ratio	Notes
2	1.04 μ s	24.4 ns	43 $\times$	2 native slots
8	1.32 μ s	17.2 ns	77 $\times$	8 native slots
32	1.10 μ s	15.9 ns	69 $\times$	$T = 32$ (native boundary)
48	2.16 μ s	29.1 ns	74 $\times$ ^a	26N + 6S + 16V (overflow update)
128	3.78 μ s	15.2 ns	249 $\times$	26N + 6S + 96V (overflow update)

^a Updating a native slot on a 48-total-slot object costs $\approx 1.0 \mu$ s (39 $\times$); the 74 $\times$ row benchmarks an overflow-slot update, illustrating the additional trie path-copying cost when the target slot lies beyond the 26 user-defined native positions.

5.1.1 Predicate Dispatch Micro-benchmark. We measured the per-invocation cost of a 20-method generic function, each clause specialized on a distinct Level-3 predicate, comparing FOL open and closed dispatch against standard CLOS type dispatch and a manual `cl:cond` cascade Table 8, SBCL 2.6.0 x86-64).

FOL `defn` (closed) and `defgeneric` (open) both add $\approx 2.0\times$ overhead, indistinguishable from the `cl:cond` baseline within noise

Table 7: Modeled update cost ($\mu\text{s}/\text{op}$) at candidate thresholds T

N	T=8	T=12	T=16	T=24	T=32	T=32 strategy
8	0.50	0.46	0.45	0.45	0.45	all-native
12	0.80	0.67	0.66	0.65	0.66	all-native
16	0.97	1.10	0.86	0.85	0.85	all-native
24	1.27	1.27	1.29	1.22	1.24	all-native
32	1.52	1.56	1.55	1.69	1.64	all-native
48	2.42	2.11	2.33	2.49	2.36	32N + 16V

Bold = minimum in row. “ $kN + jV$ ” = copy k native slots then batch-build j -element overflow vector. All-native = copy N slots directly. Simulated copy costs only; actual update-slots additionally incurs MOP iteration overhead (see Table 6).

Table 8: Predicate Dispatch Performance ($N = 20$, 5 trials $\times$ 1 M iterations).

Dispatch Method	Mean/op	Range	Ratio
Standard CLOS (Type)	31.3 ns	0.9 ns	1.00 $\times$
Manual cond Cascade	64.0 ns	2.1 ns	2.04 $\times$
FOL defn (Closed)	62.3 ns	0.4 ns	1.99 $\times$
FOL defgeneric (Open)	63.1 ns	2.1 ns	2.02 $\times$

Listing 8: Naive Persistent BFS Step

```
(defn bfs-step [q dists graph]
  (if (empty? q)
    dists
    (bind [u (first q)]
      (edges (get graph u)
        d (get dists u)]
      (loop [es edges nq (rest q) nd dists]
        (if (empty? es)
          (bfs-step nq nd graph)
          (bind [v (first es)]
            (if (= (get nd v) -1)
              (recur (rest es)
                (conj nq v)
                (assoc nd v (inc d))))
              (recur (rest es) nq nd))))))))
```

(± 2.1 ns). Open dispatch scans linearly without a cache, so all-Level-2 structures also incur an $O(N)$ penalty. Overhead scales with clause count (Table 9): at $N = 50$ the measured total is $3.6\times$ CLOS.

5.1.2 Evolution Micro-Benchmark. Migration costs $\approx 1.6\times$ CL at 1 hop. At depth, SBCL history machinery dominates; FOL is bounded at $\leq 12.2\mu\text{s}$ and converges with CL at 3 hops (Table 11).

5.2 Naive-Translation Workloads

Persistent *slot* updates are $40\text{--}250\times$ slower than mutation (Table 6). Naive BFS (Listing 8) illustrates the path-copying penalty (Table 12).

Transients cut allocation significantly and improve BFS overhead from $41.7\times$ to $6.0\times$.

Idiomatic Reformulations. Algorithmic restructuring eliminates the translation penalty (Table 13). **Lazy BFS** initializes nodes on discovery, bypassing N initial assoc calls.

Lazy BFS trims initialization waste but its per-edge checks make it slower ($18.5\times$) than naive pre-allocated transients ($6.0\times$).

5.3 Abstract Syntax Tree Rewriting Engine

We implemented an AST optimization engine (25 node classes, 54 rewrite rules via predicate dispatch with nested patterns, 25-clause walk) to evaluate both contributions together.

```
(defmethod ast-optimize
  [({:keys [(left (:keys [(val (eq1 0))]) <op-lit>])
    (right r)])] <op-add>])
  r)
```

The built-in `eq1` specialization (`(val (eq1 0))`) expands to `(lambda (x) (eq1 x 0))`, matching a slot against a specific value; it appears in the grammar’s predicate position. The CL baseline uses `defstruct` with `typecase` on a 9,999-node balanced tree—`defstruct` is the fastest idiomatic CL baseline, giving the *worst-case* overhead estimate for FOL. Practitioners migrating `defclass`-based code should consult Section 5.5 for the $5.5\times$ adoption-cost estimate under the realistic `defclass/defmethod` baseline.

FOL is $202\times$ slower per pass (35 GC cycles vs. 1 for CL); the $116\times$ build ratio exceeds micro-benchmarks due to per-node `initialize-instance` overhead. Transient Builders mitigate this via mutable construction before freezing.

5.4 Logic Simulation (LSim)

LSim is an event-driven digital logic simulator. The CL baseline uses a mutable destructive leftist heap [8] with mutable hash-table gate state; **FOL** uses a persistent leftist heap [23] with HAMT gate state. A 2×2 ablation (Table 16) isolates component contributions. Table 15 covers six configurations; the largest circuits used a 56 GB heap.

At scale, the CL baseline allocates one `make-hash-table` per gate ($32,768$ tables at $32\times 32\times 32$); as the simulation runs, these objects scatter across the heap and degrade GC scan locality, causing CL’s $4.5\times$ per-gate-eval slowdown between the $8\times 32\times 32$ and $32\times 32\times 32$ configurations. FOL’s HAMT gate-state entries are depth-bounded nursery allocations collected before reaching the old generation; this GC-locality property—not persistence per se—drives the $3.8\times$ advantage. This advantage is specific to the per-gate hash-table allocation pattern: a CL baseline using a single shared hash table keyed by `(gate, slot)` pairs might not exhibit the same fragmentation. A 2×2 ablation (Table 16) breaks down these costs.

FOL matches Hybrid-A at $32\times 32\times 32\text{--}30000$: wall-times ranged from $51.8\text{k--}52.3\text{ k s}$ for CL, and $13.4\text{k--}13.6\text{ k s}$ for FOL. Differences are within $\pm 110\text{ s}$ noise. Hybrid-B falls from $6.3\times$ to $1.35\times$; as scale grows, the CL baseline itself severely degrades from hash-table fragmentation, making Hybrid-B’s fixed $O(\log n)$ heap tax look comparatively better. The $3.8\times$ advantage stems from HAMTs avoiding heap fragmentation.

5.5 CLOS Adoption Cost: Expression Interpreter

We port an idiomatic CLOS expression interpreter (seven node types, three generic functions: `eval-expr`, `pretty`, `free-vars`) to FOL with minimal structural changes. Unlike the AST optimizer

Table 9: Dispatch Scaling: mean time/op (ns) at N clauses (5 trials $\times$ 1 M iterations). Ratios vs. CLOS. CLOS is $O(1)$ (type-indexed cache); predicate scan methods are $O(N)$.

N	CLOS (ns)	COND (ns)	FOL defn (ns)	FOL defgen (ns)	COND/ $\times$	defn/ $\times$	defgen/ $\times$
5	28.7	41.0	39.0	40.2	1.43	1.36	1.40
10	30.0	48.6	46.9	47.8	1.62	1.57	1.60
20	31.3	64.0	62.3	63.1	2.04	1.99	2.02
50	28.7	105.8	104.1	104.6	3.68	3.62	3.64

Table 10: Single-hop lazy migration: FOL vs. CL (μ s/op, 10K instances)

Phase	FOL	CL	Notes
Migration (first touch)	11.99	7.33	one-time per dormant instance
of which:	10.62	7.31	SBCL migration + alias recovery
overhead vs. steady-state			
Steady-state update	1.37	0.02	post-migration, repeated use

Table 11: Multi-hop lazy migration first-touch cost (μ s/op, 5K instances per cohort)

Hop depth	FOL	CL	Notes
1-hop (v2 $\rightarrow$ v3)	12.18 ^b	4.10	FOL allocates new object; CL mutates in-place
2-hop (v1 $\rightarrow$ v3)	7.59	5.62	FOL’s single-pass fold maintains $O(A)$ theoretical scaling.
3-hop (v0 $\rightarrow$ v3)	5.38	5.13	SBCL migration machinery dominates; FOL and CL costs converge.
Steady-state	1.09	0.01	post-migration; uniform across all cohorts

^b Higher than Table 10’s 11.99 μ s: instances created at v2 carry two prior class redefinitions in SBCL’s version history, incurring more migration overhead than instances created at v1 or v0 regardless of FOL hop depth. This explains the apparent anomaly that 1-hop costs more than 2-hop.

Table 12: Naive-Translation Workload Results

Workload	Implementation	Time	Alloc	vs. CL
BFS	CL (hash-table)	4 ms	2.3 MB	1.0 $\times$
$N = 50,000$	FOL Persistent (assoc)	163 ms	119 MB	41.7 $\times$
	FOL Transient (assoc!)	24 ms	13 MB	6.0 $\times$

Table 13: Idiomatic Algorithm Results: persistent and transient FOL vs. native CL

Workload	Implementation	Time	Alloc	vs. CL
BFS lazy	CL (hash-table)	3 ms	2.3 MB	1.0 $\times$
$N = 50,000$	FOL persistent (lazy dict)	125 ms	61 MB	37.8 $\times$
	FOL transient (hamt-assoc!)	61 ms	16 MB	18.5 $\times$

Table 14: AST Optimizer: FOL vs. CL ($N = 100$ passes, 9,999-node balanced tree).

Phase	Implementation	Time	Alloc	vs. CL
Build	CL	0.516 ms	0.16 MB	1.0 $\times$
	FOL	60.1 ms	23.95 MB	116 $\times$
Optimizer (per pass)	CL	0.357 ms/pass	0.125 MB/pass	1.0 $\times$
	FOL	72.1 ms/pass	17.91 MB/pass	202 $\times$

Table 15: LSim PQ Scaling Results (mean wall time).

Config	Gate Evals	CL	FOL	Ratio (FOL/CL)
8bit-100	876	78 ms	109 ms	1.4 $\times$
32bit-300	10 224	94 ms	141 ms	1.5 $\times$
8x32-900	281 248	359 ms	1.87 s	5.2 $\times$
32x32-3000	3 850 304	7.57 s	34.15 s	4.5 $\times$
8x32x32-9000	89 708 032	906 s	983 s	1.1 $\times$
32x32x32-30000	1,183,510,528	52,050 s	13,531 s	0.26 $\times$ ^a

^a Small-circuit times include shared ≈ 74 ms initialization overhead; ratio < 1 means FOL is faster. The largest config was ablated (Table 16); the fragmentation explanation is supported by the 2 $\times$ 2 breakdown where persistent maps avoid hash-table degradation.

Table 16: 2 $\times$ 2 Ablation: Mutable vs. Persistent Heap $\times$ State Maps (mean wall time; ratios relative to CL baseline). Hybrid-A = persistent maps + mutable heap; Hybrid-B = mutable maps + persistent heap; FOL = both persistent.

Circuit	CL	Hybrid-A	Hybrid-B	FOL
32bit-300	94 ms	141 ms (1.5 $\times$)	125 ms (1.3 $\times$)	141 ms (1.5 $\times$)
8x32-900	359 ms	1.87 s (5.2 $\times$)	2.25 s (6.3 $\times$)	1.87 s (5.2 $\times$)
32x32-3000	7.57 s	34.1 s (4.5 $\times$)	42.7 s (5.6 $\times$)	34.1 s (4.5 $\times$)
8x32x32-9000	906 s	983 s (1.1 $\times$)	1646 s (1.8 $\times$)	983 s (1.1 $\times$)
32x32x32-30000	52.0k s	13.5k s (0.26 $\times$)	70.2k s (1.35 $\times$)	13.5k s (0.26 $\times$)

(Section 5.3), which uses defstruct as the fastest-possible CL baseline, this benchmark uses defclass/defmethod—the natural starting point for a developer adopting FOL from an existing CLOS codebase—giving a more conservative and realistic adoption-cost estimate. The CL baseline uses mutable environments; FOL uses (assoc env bvar v); method bodies are otherwise identical.

A corpus of 50 depth-5 expression trees is evaluated 1,000 times (150,000 root calls across the three generic functions; up to ≈ 9.4 M recursive dispatches).

Table 17: Expression Interpreter Benchmark (SBCL 2.6.0, AMD Ryzen 9 5900X).

Phase	CL	FOL	Time ratio	Mem ratio
Build (50 trees, depth 5)	20.1 ms / 3.8 MB	35.9 ms / 8.9 MB	1.8×	2.3×
Eval (1000 iter × 50 exprs)	0.50 s / 224.5 MB	2.76 s / 1300 MB	5.5×	5.8×

Results. The 5.5× eval ratio reflects a make-instance baseline, fewer clauses, and Level-2 specializers using fast typep checks. The port requires no architectural logic shifts; differences lie purely in immutable mappings.

6 Threats to Validity

Single implementation. All measurements use SBCL 2.6.0; results may differ on other CL implementations.

Experimenter bias. FOL and all baselines share the same author; more aggressive CL tuning might narrow results.

Benchmark coverage. Smaller LSim configs were run repeatedly; the largest circuit was run three times per implementation due to its scale (≈ 14 hours per run).

Benchmark selection bias. Macro-benchmarks may over-represent FOL’s advantages; the interpreter and naive-translation benchmarks offer a more conservative view.

Interoperability hazard. CL code extending FOL classes may encounter runtime mutation errors unsignaled at definition time.

Non-portability. persistent-class uses AMOP hooks (allocate-instance, initialize-instance) but reads SBCL internals. Micro-benchmarks show 1.6× speedup (16.2 vs. 25.7 ns) over portable slot-value-using-class.

Re-evaluation and Preference Resolution. Interactive re-evaluation reorders incomparable clauses; ordering may differ between sessions. prefer-method (Section 4) is not yet implemented; when available, it will eliminate load-order dependence. Current workarounds require careful clause ordering or restructuring; implementation is planned for near-term release.

All-Level-2. Type-only defmethod is expected to incur the same $O(N)$ scan as predicate dispatch, but this is not empirically verified.

7 Related Work

Persistent data structures. FOL’s trie follows Bagwell’s HAMT [3] and Clojure [16], using path-copying persistence [10]; Okasaki’s heaps [23] underpin our LSim heap. Using these structures, FOL extends CLOS to persistent *objects* with predicate dispatch.

Schema evolution. Gemstone [25] and ENCORE [26] pioneered lazy migration, but require explicit coercion code for renames [2]. FOL provides declarative migration via :alias; multi-hop chains compose in a single pass via the CLOS MOP. This model is closer in spirit to Datomic’s attribute renames, where identity is decoupled from specific attribute keys.

OOP/FP bridges and pattern dispatch in other languages. Scala’s case classes [22] and match [11] provide immutable ADTs with structural dispatch, but updates require explicit copy and the class

system is not MOP-integrated. Racket [13] separates match from its class system [14]; Julia [4] and MultiJava [7] provide multi-parameter type dispatch without predicate specializers.

Predicate dispatch. JPred [12] requires a theorem prover for ordering. Millstein [21] achieves decidability by restricting predicates to linear arithmetic; FOL instead admits arbitrary predicates via syntactic leveling, trading decidable subsumption for an open language. Later Cecil work [6] achieves efficient dispatch via offline decision-tree compilation. Our composite level recovers ranking for type-annotated predicates statically. GHC’s overlapping instances [15] order heads via pragmas but lack a predicate stratum or evolution. Filtered dispatch [24] introduced guard predicates. Racket’s match [13] provides structural matching but is decoupled from its class system. CL libraries like trivia offer pattern matching but target control flow rather than open generic dispatch. Clojure’s defmulti [16] uses user-supplied functions and prefer-method; protocols [18] dispatch on the first argument’s class. FOL’s defmethod occupies defmulti’s niche but adds predicate specializers and CLOS method combinations, trading an $O(N)$ scan for expressivity.

8 Conclusion

FOL contributes a persistent metaclass system and specificity-ordered predicate dispatch. Hybrid storage ($T = 32$) keeps reads fast while lazy migration enables transparent evolution. Specificity ordering handles structurally differentiated patterns, confining load-order sensitivity to structural ties.

Slot updates are 40–250× slower; FOL is a scale-out solution for complex architectures rather than a latency-optimized replacement for mutable slots. The dichotomy between the logic simulator’s scale advantage and the AST rewrites’ penalty provides a heuristic: persistent objects excel for “macro” structures (large maps, sparse graphs) but are inappropriate for fine-grained pointer structures due to GC pressure. Future work includes dispatch caching and static analysis for unreachable clause detection.

FOL is publicly available [1].

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